A Controlled Study of Pure Monte Carlo Tree Search for Symbolic Regression

After conversations with GPT

August 14, 2026

1 Setup

1.1 Problem

Data $(x_i, y_i)_{i=1}^n$ are generated by a fixed unknown expression h, $y_i = h(x_i)$. Symbolic regression searches a discrete set Π of expression *structures*; a structure $\pi \in \Pi$ carries free coefficients β, fitted by least squares,

$$\hat{\beta}_\pi = \arg\min_\beta \frac{1}{n} \sum_{i=1}^n \big(y_i - \pi_\beta(x_i)\big)^2,$$

and is scored by the coefficient of determination of that fit, clipped to $[-1, 1]$ and written $R(\pi)$ (6). The clip is load-bearing rather than cosmetic: fitting βx to `lin_C` of Table 2 returns $R^2 \approx -2.9 \times 10^{12}$, and one such completion would otherwise decide the mean value of a whole subtree. The inner fit is part of the score, so R is a function of the structure alone and the symbolic search is $\pi^* = \arg\max_{\pi\in\Pi} R(\pi)$. The motivating instance is a quadratic target $h(x) = c_0 + c_1 x + c_2 x^2$, with structures ranging over the subsets of $\{1, x, x^2\}$.

We deliberately strip the problem to isolate the search mechanism: no variable-length trees, no invalid expressions, no noise, no complexity penalty, no additional operators, and no neural policy or value network. The structure search becomes *subset selection*. The purpose of the study is to characterize the finite-budget limitations of pure Monte Carlo tree search (MCTS) in grammar-based symbolic regression. The neural-guidance question is scientifically separate: before asking whether a network improves search, one must determine what information the underlying pure-search procedure estimates and when that information is aligned with symbolic optimization.

1.2 Values and failure modes

A state s is a partial derivation, $\mathcal{A}(s)$ its set of legal actions, and $T : \mathcal{S} \times \mathcal{A} \to \mathcal{S}$ the deterministic transition. Let

$$\mathcal{E}(s) = \{\pi : \pi \text{ is a terminal expression completable from } s\},$$

so that $\Pi = \mathcal{E}(s_0)$ is the structure set of §1 and a terminal expression π is itself a state, with $\mathcal{E}(\pi) = \{\pi\}$. The best-achievable value is

$$V^*(s) = \max_{\pi\in\mathcal{E}(s)} R(\pi), \tag{1}$$

where $R(\pi)$ is the terminal symbolic-regression reward. The uniform completion policy

$$q(a \mid s) = \frac{1}{|\mathcal{A}(s)|}, \qquad a \in \mathcal{A}(s),$$

induces, because transitions are deterministic, a distribution $q(\cdot \mid s) \in \Delta(\mathcal{E}(s))$ over completions; the mean-rollout value is

$$V^q(s) = \mathbb{E}_{\pi\sim q(\cdot|s)}\big[R(\pi)\big]. \tag{2}$$

Every target below is exactly expressible in the grammar, so $V^*(s_0) = 1$, and the exact-completion mass is

$$\rho(s) = \Pr_{\pi\sim q(\cdot|s)}\big[R(\pi) \geq 1-\tau\big], \qquad \tau = 10^{-6}, \tag{3}$$

the probability that a uniform completion of s fits the target exactly. The threshold is the *global* optimum $V^*(s_0) = 1$, not the local $V^*(s)$, so it is the same at every state and ρ obeys the same backward induction as V^q. A single tolerance τ serves here and in the exact-recovery indicator (23); on the six targets that carry the study every exact fit returns $R = 1$ to the last bit, so no reported value of ρ depends on it (Remark 3.3).

Question 1.1. Which mechanisms make pure MCTS fail at finite budget?

Two candidate mechanisms:

1. *Value-semantic failure*: for sibling states s_1, s_2,

$$V^q(s_1) > V^q(s_2) \quad \text{while} \quad V^*(s_1) < V^*(s_2).$$

   Mean-based search then confidently exploits the wrong branch: a shallow decoy yields a high expected reward, while the branch containing the true optimum is penalized by its abundance of poor completions.

2. *Rare-event failure*: $\rho(s) \ll B^{-1}$ at budget B. Search may explore the correct branch yet exhaust the budget before sampling a near-optimal leaf.

Therefore $V^q(s)$, $V^*(s)$ and $\rho(s)$ are the three primary metrics of every experiment below. Table 1 summarizes the two mechanisms.

Failure mechanism	**Formal condition**	**Intuitive cause**	**Implication for search**
Value-semantic failure	$V^q(s_1) > V^q(s_2)$ while $V^*(s_1) < V^*(s_2)$ for sibling states s_1, s_2	A suboptimal branch has the higher mean rollout reward: a shallow decoy looks better	Search confidently exploits the wrong branch and starves the branch containing the true global optimum.
Rare-event failure	$\rho(s) \ll B^{-1}$	Near-optimal completions carry a negligible fraction of the rollout probability mass.	Search explores the correct branch but exhausts the budget B before sampling a near-optimal leaf.

Table 1: The two finite-budget failure mechanisms of mean-based MCTS.

Remark 1.2. The optimization objective depends on V^*, whereas conventional UCT uses empirical mean returns as its exploitation statistic [6]; asymptotic consistency does not imply effective finite-budget search, and adverse finite-depth behavior is known for UCT [3]. The goal in symbolic regression is one exceptional expression, not a policy with high average return: best-focused Monte Carlo methods [1,2], risk-seeking symbolic regression [8], and recent MCTS systems with learned proposals or extreme-value allocation [4,5] all respond to this mismatch.

Experiment 1 measures these quantities exactly on a controlled family. Experiment 2 tests whether they predict the finite-budget behavior of pure MCTS.

1.3 Notation

Three families of symbol are easy to confuse and are kept apart throughout: math italic c_p for the coefficients of the *target*, typewriter `C0`, `C1`, `C2` for the *grammar tokens* whose values are fitted, and C for a distinguished *state*.

Symbol	Type	Meaning
$s \in \mathcal{S}$	state	sentential form in the length-L buffer
s_0	state	root, the one-token form (S)
$\pi \in \Pi$	state	terminal expression; $\Pi = \mathcal{E}(s_0)$
$\mathcal{E}(s)$	set of states	terminals completable from s
$a = \mathrm{pos} \cdot P + \mathrm{prod}$	action	flattened $(\mathrm{pos}, \mathrm{prod})$ index
$\mathcal{A}(s)$	set of actions	the legal mask at s, (5)
$T(s, a)$	map $\mathcal{S} \times \mathcal{A} \to \mathcal{S}$	deterministic transition
$w_0, \ldots, w_{P-1}$	token strings	the productions, in code order
S, A	nonterminals of G	italic; calligraphic $\mathcal{S}, \mathcal{A}$ are the state and action sets
$R(\pi)$	$\mathbb{R}$	$\mathrm{clip}(R^2, -1, 1)$ after the inner fit, (6)
V^*, V^q, ρ	$\mathcal{S} \to \mathbb{R}$	(1), (2), (3)
C, D_1, D_2, D_3	states	the four children of s_0, Section 3.1
c_0, c_1, c_2	$\mathbb{R}$	the target's chosen coefficients, (7)
`C0`,`C1`,`C2`	grammar tokens	free constants fitted by `lmfit`
B, K	$\mathbb{N}$	terminal-evaluation budget; independent completions
τ	$\mathbb{R}$	exactness tolerance 10^{-6}, (3)

The reachable states form a directed acyclic graph rather than a tree: an action names a buffer position, so expanding the two nonterminals of `+ S S` in either order reaches the same form. Experiment 1 therefore runs a backward induction over 4,898 *states*, whereas the search tree Experiment 2 builds is a tree over *action sequences*, in which one form may occupy several nodes.

2 The derivation MDP in the code

Definition 2.1 (The grammar-derivation MDP)**.** Fix a context-free grammar G with start symbol S, production set $\{w_0, \ldots, w_{P-1}\}$ indexed in code order, and a buffer length L (`max_len`). The environment is the deterministic undiscounted MDP:

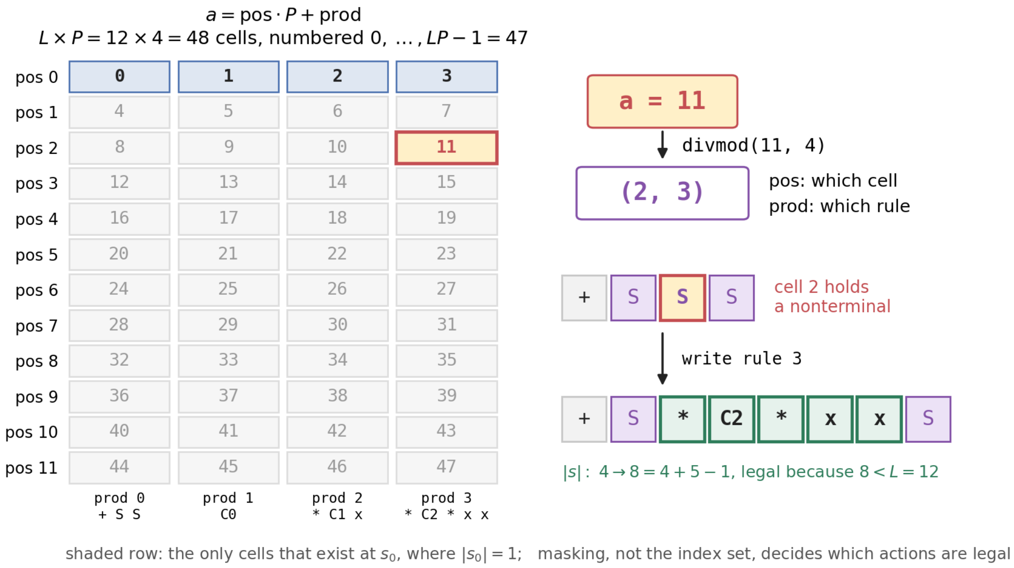
- *State.* A sentential form $s \in \Sigma^*$ with $|s| \le L$, stored as a fixed-length array of L tokens together with $|s| =$ `real_state_len`; cells beyond $|s|$ hold a pad token. The observation space is `MultiDiscrete`$([\,|\Sigma| + 1\,]^L)$, so the pad symbol is observable and the agent sees how much buffer remains.

- *Start.* $s_0 = (S)$, a single nonterminal, $|s_0| = 1$.

- *Action.* An action must say *which* cell of the buffer to rewrite and *which* production to write there, so the action set is the set of pairs $(\mathrm{pos}, \mathrm{prod}) \in \{0, \ldots, L-1\} \times \{0, \ldots, P-1\}$. There are LP such pairs, and `Discrete(L*P)` indexes them by one integer counted from zero, whence

$$a \in \{0, \ldots, LP - 1\}, \qquad a = \mathrm{pos} \cdot P + \mathrm{prod}, \qquad (\mathrm{pos}, \mathrm{prod}) = \mathrm{divmod}(a, P). \tag{4}$$

  The -1 is zero-indexing of LP cells and nothing more; at $L = 12$, $P = 4$ the largest action is 47.



- *Legal set.* $a \in \mathcal{A}(s)$ iff

$$\text{pos} < |s|, \quad s[\text{pos}] \text{ is a nonterminal}, \quad \text{prod} \in \text{proddict}[s[\text{pos}]], \quad |s|+|w_{\text{prod}}|-1 < L. \tag{5}$$

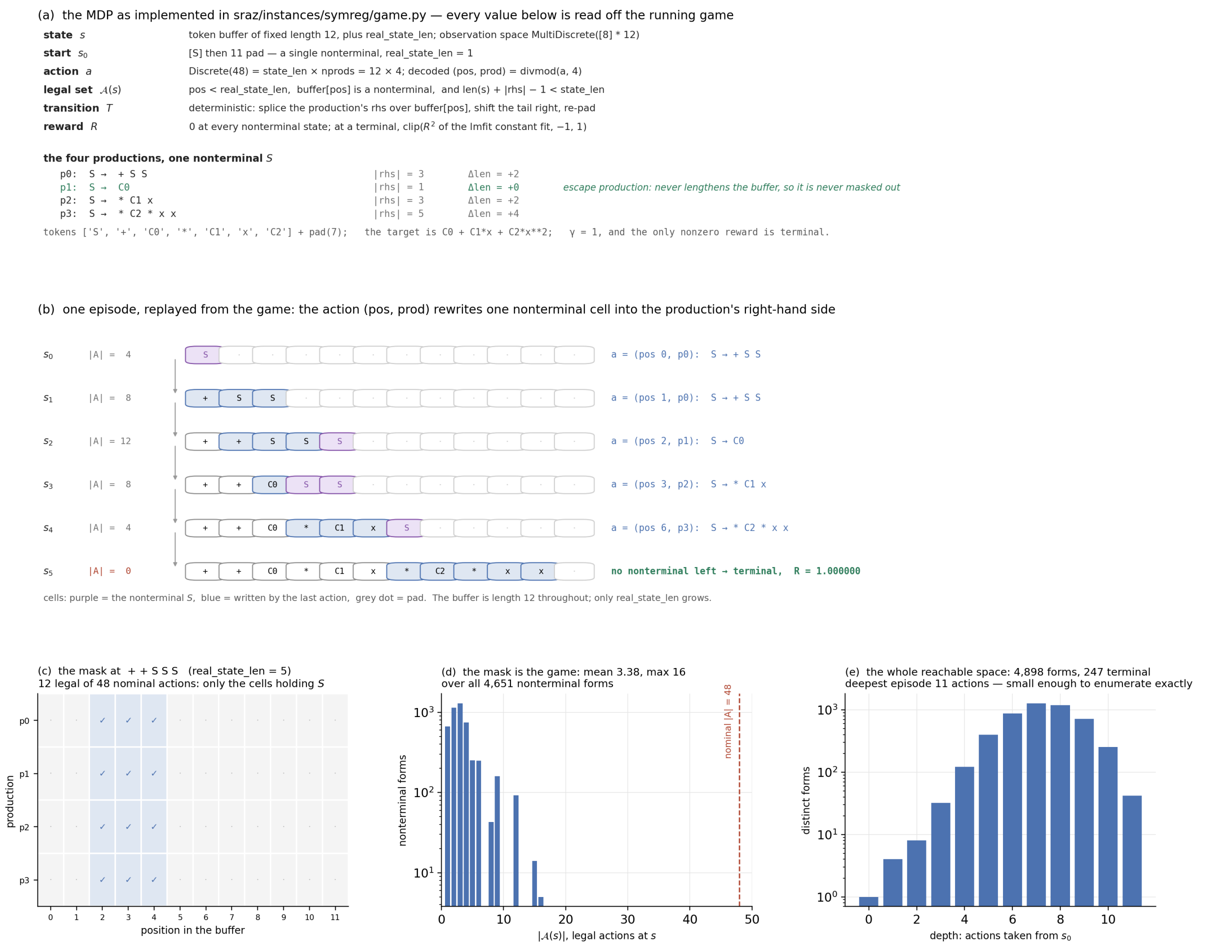
- *Transition.* $T(s, a)$ splices w_{prod} over cell pos, shifts the tail right, and re-pads. It is a function, not a kernel.

- *Reward.* $R \equiv 0$ at every nonterminal state. At a terminal, the token string is read as a prefix expression, its `C*` tokens are fitted by `lmfit`, and the reward is the clipped coefficient of determination

$$R(\pi) = \text{clip}\big(R^2, -1, 1\big), \tag{6}$$

with -1 returned for any unparseable expression or failed solve.

- *Discount.* $\gamma = 1$; exactly one reward per episode is nonzero, so the return of an episode is its terminal R^2.



Two features of (4)–(5) matter for everything downstream. First, the action space is nominally LP but almost entirely illegal: on the additive instance at $L = 12$, $LP = 48$ while the legal set averages 3.38 and never exceeds 16 over all 4,651 nonterminal forms. The mask, not the action space, is what search branches on. Second, because an action names a position, the uniform completion policy q of (2) is uniform over the flattened $(\mathrm{pos}, \mathrm{prod})$ mask, *not* uniform over the productions of one designated nonterminal. The two measures differ as soon as a form holds more than one nonterminal, and V^q inherits whichever is used, so which one is meant must be stated.

2.1 Termination

The environment has no step limit, no truncation, and no stop action. An episode ends when the sentential form runs out of nonterminals, and not before. Concretely, `step` has three exits, and only the first can fire under masked play.

Exit 1: normal termination. `terminated` $= \neg$`_has_nonterms`(s). The finished string is decoded, converted to infix, fitted, and scored by (6). This is how every episode in every experiment ends.

Exit 2: invalid action. A defensive branch returns $R = -1$ with `terminated` = true when the decoded action is out of range, addresses a terminal cell, or would overflow the buffer. Calling `step` off-mask on s_0 reproduces it exactly. Under masked play it is unreachable; it exists so that a masking bug fails loudly rather than looping.

Exit 3: dead end. A form retains a nonterminal but $\mathcal{A}(s) = \emptyset$. Then `step` is never called at all: `MCTS._rollout_value` breaks out of its loop and *discards* the sample rather than scoring it. Proposition 2.3 shows this exit cannot fire.

Proposition 2.2 (Length bound)**.** Every reachable form satisfies $|s| \le L - 1$. In particular every terminal expression has at most $L - 1$ tokens.

Proof. Induction on the number of actions. $|s_0| = 1 \le L - 1$. If s' follows s by a legal action with production w, the mask condition $|s| + |w| - 1 < L$ in (5) states exactly that $|s'| < L$, hence $|s'| \le L - 1$. $\square$

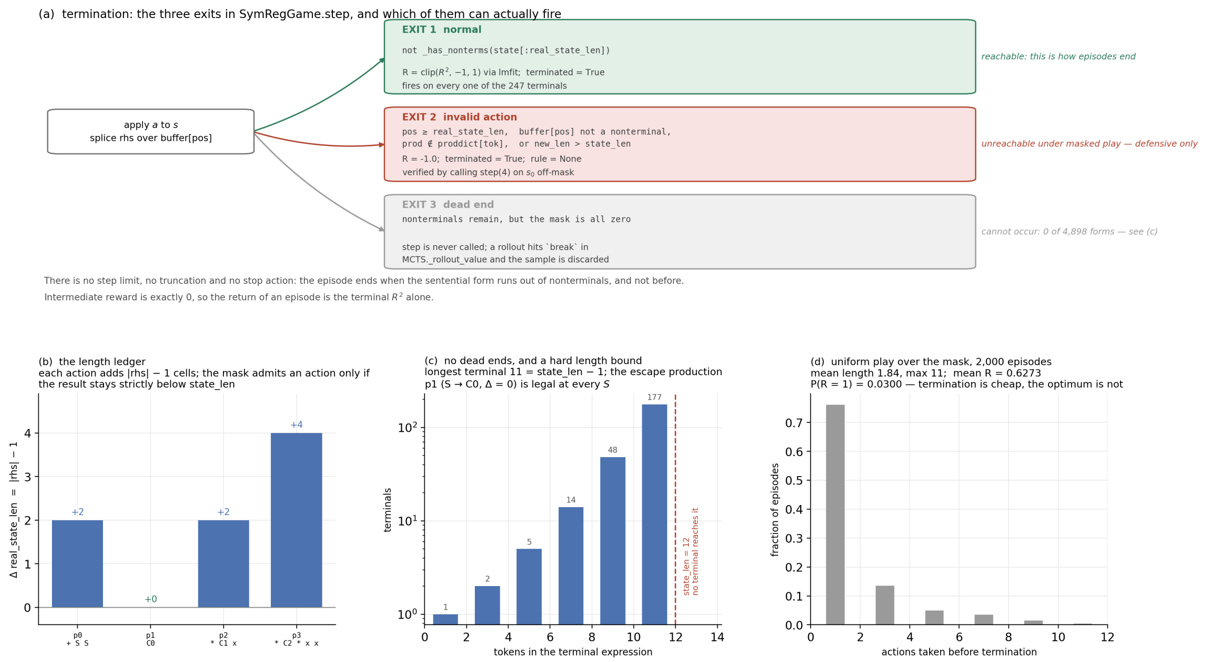
The bound is attained, and the strict inequality in (5) is load-bearing: at $L = 12$ the longest terminals have 11 tokens, and the exact expression `+ C0 + * C1 x * C2 * x x` is admissible so that $V^*(s_0) = 1$. At $L = 11$ it is excluded, and $V^*(s_0)$ then depends on the target: it stays at 1 for every linear target, which needs only five tokens, but falls to 0.997213 on `quad_B` and 0.527856 on `quad_A` (Section 3). A buffer must be set one longer than the longest expression one intends to be reachable.

Proposition 2.3 (No dead ends)**.** Suppose every nonterminal A of G admits a production $A \to w$ with $|w| = 1$, an *escape production.* Then $\mathcal{A}(s) \neq \emptyset$ for every reachable s containing a nonterminal, so Exit 3 never fires.

Proof. Let $s[i] = A$ be a nonterminal and $A \to w$ an escape production. The first three conditions of (5) hold at that pair by construction. For the fourth, $|s| + |w| - 1 = |s| \le L - 1 < L$ by Proposition 2.2. The action is legal, so the mask is nonempty. $\square$

Both shipped grammars satisfy the hypothesis through $S \to$ `C0`; an exhaustive census finds 0 dead ends among the 4,898 reachable forms. The hypothesis is a genuine constraint rather than a formality. Removing the constant production, or capping L below the shortest completion of some reachable form, reintroduces dead ends — and since the rollout discards them instead of scoring them, they would bias V^q silently rather than raise an error.

Remark 2.4 (Parity)**.** Each action changes the length by $|w| - 1 \in \{0, 2, 4\}$, all even, and $|s_0| = 1$. Every reachable form therefore has odd length, and terminals can only hold $1, 3, 5, 7, 9, 11$ tokens. The census agrees exactly: $1, 2, 5, 14, 48, 177$ terminals at those six lengths.



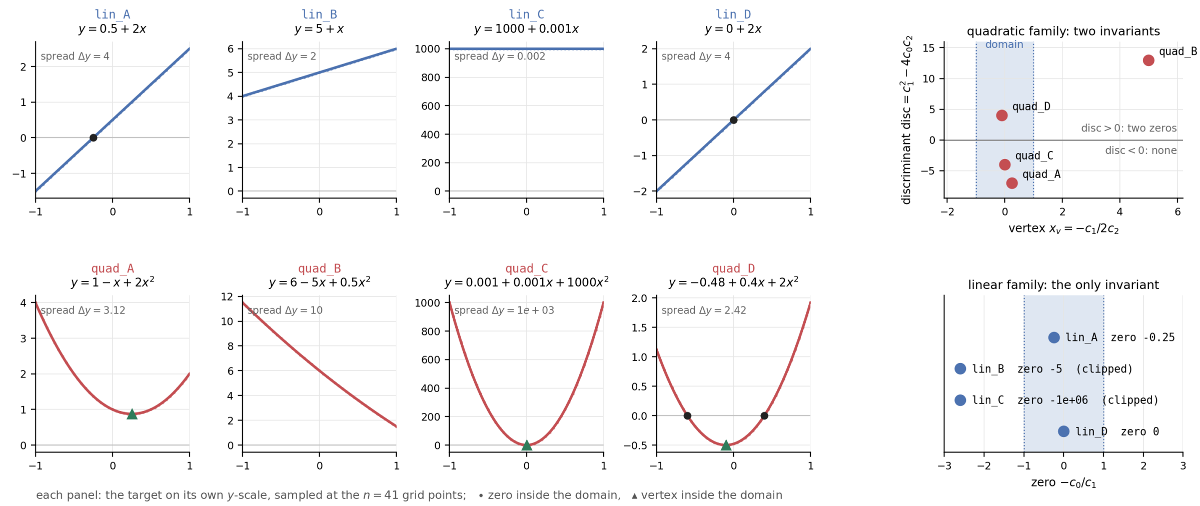
3 A minimal controlled family

The MDP of Definition 2.1 is now held fixed — grammar `ADDITIVE_GRAMMAR`, buffer $L = 12$ — and the target the terminal expression is fitted against is the only thing that varies.

Definition 3.1 (Target family)**.** A *target* is a polynomial together with the grid it is sampled on,

$$y(x) = \sum_{p=0}^{d} c_p\, x^p, \qquad x_1, \ldots, x_n \text{ equally spaced in } [-1, 1], \quad n = 41, \tag{7}$$

with $d \leq 2$ and coefficients *chosen*, not drawn. The family has eight members: four linear, separated by where the zero $-c_0/c_1$ falls relative to the domain, and four quadratic, separated by the vertex $x_v = -c_1/2c_2$ and the discriminant $\mathrm{disc} = c_1^2 - 4c_0c_2$.[1]



Each member is a distinct shape over the domain — monotone or turning, crossing zero or not — and two of them spread their coefficients over six orders of magnitude (`lin_C`, `quad_C`) as a probe of the inner least-squares solve. The solve survives it to eleven digits; what does not survive is the exactness test, so those two are reported and then set aside (Remark 3.3). Every member is exactly expressible in the grammar, so $V^*(s_0) = 1$ throughout and any shortfall is the search's, not the grammar's.

3.1 Where the difficulty is

The decoys are already in the grammar. Exactly $|\mathcal{A}(s_0)| = 4$ actions are legal at the root, one per production, and they have four distinct successors. We name those four successor *states* after the production that creates them,

$$D_1 = \texttt{C0}, \qquad D_2 = \texttt{* C1 x}, \qquad D_3 = \texttt{* C2 * x x}, \qquad C = \texttt{+ S S}, \tag{8}$$

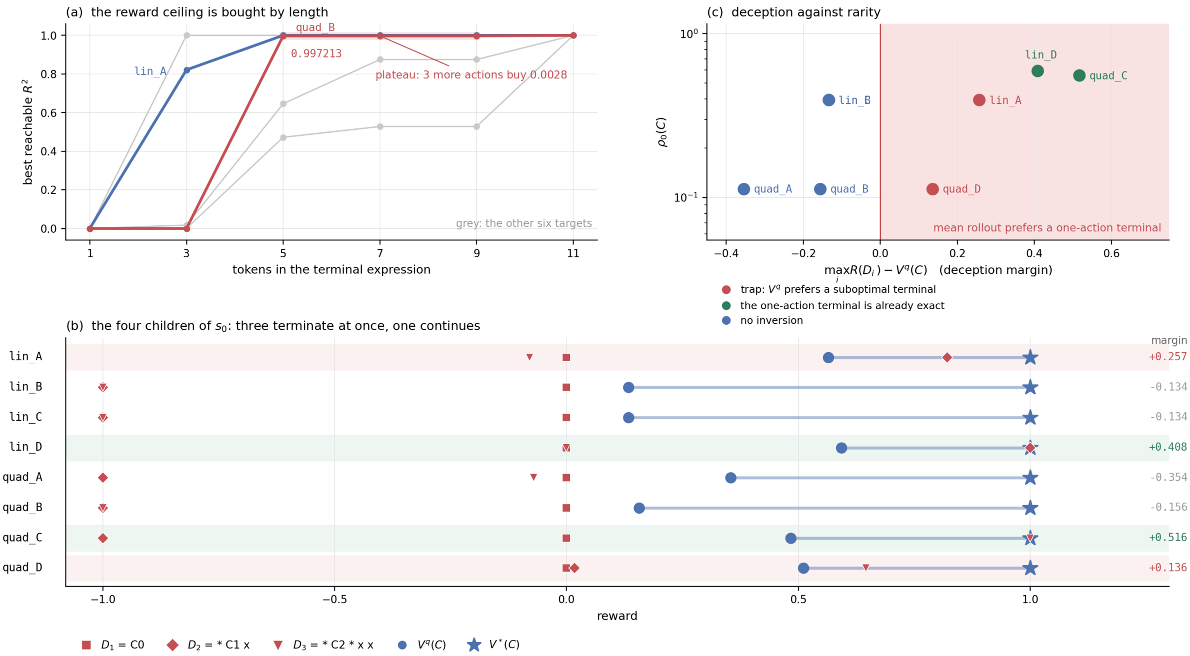
so that $D_i, C \in \mathcal{S}$ and every value written against them — $V^*(C)$, $V^q(C)$, $\rho(C)$, $R(D_i)$ — is the state function of §1.2 evaluated at that state. Because the four root actions are in bijection with these four children, we also write $N(s_0, C)$ for the visit count of the edge $s_0 \to C$; no separate symbol for a root action is needed.

[1] x_v says where the curve turns and disc whether it meets the axis at all (disc > 0 two real zeros, < 0 none), so the pair fixes the shape on $[-1, 1]$: of the four members only `quad_B` is monotone there, and only `quad_D` crosses zero inside the domain. The word *root* is reserved throughout for the MDP root s_0.

Three of the four productions carry no nonterminal on the right, so D_1, D_2 and D_3 are already terminal: each is a one-token derivation with $V^*(D_i) = V^q(D_i) = R(D_i)$, one deterministic number. The fourth child C is the only one that continues, and the only one whose subtree contains an exact expression: $V^*(C) = 1$ for every target. Root value-semantic failure is therefore a single inequality,

$$\max_i R(D_i) > V^q(C) \qquad \text{while} \qquad \max_i R(D_i) < V^*(C) = 1, \tag{9}$$

and nothing has to be added to the environment to produce it.



Both sides of (9) are tabulated in Table 2. Only `lin_A` and `quad_D` satisfy it. For `lin_D` and `quad_C` the margin $\max_i R(D_i) - V^q(C)$ is also positive, but there the one-action terminal is itself exact — for `quad_C` only to within round-off, Remark 3.3 — so preferring it is correct and no trap exists. A positive margin alone is not deception; the second inequality of (9) is what separates the two cases.

Target	$y(x)$	$R(D_1)$	$R(D_2)$	$R(D_3)$	$V^q(C)$	$\rho(C)$	margin	
`lin_A`	$0.5 + 2x$	0	0.8214	-0.0793	0.5646	0.3945	$+0.2569$	trap
`lin_B`	$5 + x$	0	-1	-1	0.1341	0.3945	-0.1341	—
`lin_C`	$1000 + 0.001x$	0	-1	-1	0.1341	0.3945	-0.1341	—
`lin_D`	$2x$	0	1	0	0.5924	0.5924	$+0.4076$	solved at once
`quad_A`	$1 - x + 2x^2$	0	-1	-0.0711	0.3540	0.1128	-0.3540	—
`quad_B`	$6 - 5x + 0.5x^2$	0	-1	-1	0.1563	0.1128	-0.1563	plateau
`quad_C`	$0.001 + 0.001x + 1000x^2$	0	-1	1	0.4840	0.5569	$+0.5160$	solved at once
`quad_D`	$-0.48 + 0.4x + 2x^2$	0	0.0170	0.6461	0.5106	0.1128	$+0.1355$	trap

Table 2: Exact root quantities for the eight targets under one fixed MDP; $V^*(C) = 1$ in every row. "Trap" marks the rows satisfying (9). The $\rho(C)$ column is at $\tau = 10^{-6}$; it is tolerance-free except on `lin_C` and `quad_C` (Remark 3.3).

The last term is the expensive one. Panel (a) plots the reward ceiling: the best R^2 any terminal of ℓ tokens attains. For `quad_B` the ceiling is 0.997213 already at $\ell = 5$, stays there at $\ell = 7$ and $\ell = 9$, and reaches 1 only at $\ell = 11$. Three of the five actions of an optimal derivation buy 0.0028 of R^2 between them. This is the trap of (9) displaced one level down: not a decoy action at the root, but a plateau inside the continue branch, where a fitted *line* through a parabola whose vertex sits at $x_v = 5$, far outside the domain, already explains 99.7% of the variance.

Rarity is a property of the target's support. The tree is the same for every target, so the only thing that can vary is which of its terminals fit.

Lemma 3.2 (Rarity depends only on the support)**.** Let $P = \{p : c_p \neq 0\}$ be the support of the target (7). A terminal π fits it exactly iff π carries an atom for every $p \in P$. Hence ρ depends on the target only through P, and

$$\rho(C) = 0.5924, \quad 0.3945, \quad 0.1128 \qquad \text{for} \qquad P = \{1\}, \quad \{0,1\}, \quad \{0,1,2\}.$$

Proof. The three productions carrying no nonterminal evaluate to 1, x and x^2, linearly independent on the 41-point grid, and a terminal is a sum of such atoms. Least squares over the atoms present is exact iff the target lies in their span, which for a polynomial of support P holds iff every $p \in P$ is present, and is otherwise strictly inexact. The predicate $R(\pi) = 1$ therefore depends on π only through the set of powers it carries, and ρ, a uniform average of that predicate over completions (16), inherits the dependence. □

Of the 247 terminals, 146 carry both 1 and x while only 12 carry all three, so the family offers exactly two rarity levels. Since completions are independent, K of them miss with probability $(1-\rho)^K$, so Pr(at least one exact) $\geq 1-\delta$ exactly when

$$K \;\geq\; K_{1-\delta} = \left\lceil \frac{\log \delta}{\log(1-\rho)} \right\rceil, \tag{10}$$

which is 6 completions at $\rho = 0.39$ and 26 at $\rho = 0.11$, for $\delta = 0.05$. Here K counts completions drawn from one fixed state, whereas the budget B of (20) counts terminal evaluations anywhere in the tree. K is what *uniform* sampling from C needs; a tree search is bound by it in neither direction, and Section 5 measures the difference.

The four cells. Two binary contrasts are therefore available without writing a line of new environment code: whether (9) holds, and whether exact completions are common ($\rho(C) = 0.39$) or rare (0.11).

	common, $\rho(C) = 0.39$	rare, $\rho(C) = 0.11$
inversion	`lin_A`	`quad_D`
no inversion	`lin_B`	`quad_A`

`quad_B` is kept as a fifth condition: no inversion at the root, but the 0.997213 plateau three actions deep. These five targets carry the study. `lin_D` is a solved-at-once control, and the remaining two are excluded for the reason that follows.

Remark 3.3 (The two ill-conditioned members)**.** `lin_C` and `quad_C` are the only rows of Table 2 that disagree with Lemma 3.2, and both disagree at the level of double-precision round-off rather than of mathematics. `lin_C` $= 10^{-3} \cdot$ `lin_B` $+ 999.995$ is an affine image of `lin_B`; R^2 is invariant under affine maps of y for every atom set containing `C0`, and every set without one clips to -1, so the two are the same problem and their $V^q(C)$ agree to 3×10^{-11}. Its exact fits nevertheless return $R = 1 - 5.6 \times 10^{-12}$, so $\rho(C)$ collapses to 0.2817 as soon as $\tau < 5.6 \times 10^{-12}$. `quad_C` fails the other way: the single atom `* C2 * x x` returns $R = 1 - 8.1 \times 10^{-12}$, which $\tau = 10^{-6}$ reads as exact — whence $R(D_3) = 1$ and $\rho(C) = 0.5569$ in Table 2 — and which any $\tau < 8.1 \times 10^{-12}$

reads as inexact, restoring $\rho(C) = 0.1128$. The window admitting both rows is $[5.6, 8.1) \times 10^{-12}$, narrower than a factor of 1.5, so no tolerance makes both mean what they say. On the other six targets every exact fit returns $R^2 = 1$ to the last bit, and every ρ reported here is unchanged for all $\tau \in [0, 10^{-6}]$.

4 Experiment 1: exact value-semantic audit

Question 4.1. Does the mean random-completion return V^q preserve the ordering induced by the best-achievable reward V^*, and how rare (ρ) is the exact expression within each partial derivation?

Hypothesis 1. The rollout mean ranks the continue branch above every one-action terminal on `lin_B` and `quad_A`, and below one of them on `lin_A` and `quad_D`:

$$\begin{aligned} V^q(C) &> \max_i R(D_i) && \text{on } \texttt{lin\_B} \text{ and } \texttt{quad\_A}, \\ V^q(C) &< \max_i R(D_i) && \text{on } \texttt{lin\_A} \text{ and } \texttt{quad\_D}, \end{aligned} \tag{11}$$

with $V^*(C) = 1 > \max_i R(D_i)$ throughout, so the second inequality is a genuine ordering failure. Independently, the completion budget required for exact recovery scales as

$$K_{1-\delta} = \Theta\left(\frac{\log(1/\delta)}{\rho}\right) \quad \text{as } \rho \to 0. \tag{12}$$

Procedure. No stochastic tree search is used in this experiment. Enumerate the mask-reachable forms once — 4,898 of them, 247 terminal — and for each of the four cells compute exactly, at every state s,

$$V^*(s), \qquad V^q(s), \qquad \rho(s). \tag{13}$$

For a nonterminal state s, define the action selected by the mean-rollout oracle as

$$a_q(s) = \arg\max_a V^q(T(s, a)), \tag{14}$$

and define its oracle sibling regret by

$$g_q(s) = V^*(s) - V^*(T(s, a_q(s))). \tag{15}$$

The reachable set is a directed acyclic graph — every action either lengthens the form or removes a nonterminal — so all three quantities follow by one backward induction over it,

$$V^*(s) = \max_a V^*(T(s, a)), \qquad V^q(s) = \frac{1}{|\mathcal{A}(s)|} \sum_{a \in \mathcal{A}(s)} V^q(T(s, a)), \tag{16}$$

and likewise for ρ, with terminal states seeded by their reward. The fitter is called once per terminal and cached, so a cell costs 247 fits, not 4,898.

Two checks guard the census. The reward of the exact derivation `+ C0 + * C1 x * C2 * x x` must come back as 1 to fit tolerance in every cell, and the induced $V^q(s_0)$ must agree with the mean of 10^5 uniform masked rollouts to within Monte Carlo error. Failure of either is an implementation error, not a finding.

Primary outputs. Report the exact root quantities of Table 2 — $R(D_i)$, $V^*(C)$, $V^q(C)$, $\rho(C)$, $K_{0.95}$ — and plot:

1. the terminal reward distribution over the 247 terminals, per cell;
2. $V^q(s)$ against $V^*(s)$, stratified by $|s|$;
3. sibling regret $g_q(s)$ against $|s|$;
4. the exact hit curve $1-(1-\rho)^K$.

Because Experiment 1 is an exact census, it requires no random seeds, confidence intervals, or significance tests.

5 Experiment 2: finite-budget pure-MCTS limits

Question. At matched terminal-evaluation budgets, do the value-order mismatch and exact-completion rarity identified in Experiment 1 predict the behavior of actual pure search?

5.1 Search conditions

We compare three conditions.

Uniform random search. Sample complete derivations independently from the uniform grammar policy and return the best terminal expression observed.

Mean-UCT. At node s, select

$$a_t = \arg\max_a \left[\overline{Q}_t(s,a) + \sqrt{\frac{2\log N_t(s)}{N_t(s,a)}} \right], \tag{17}$$

where $N_t(s)$ and $N_t(s,a)$ count the simulations through s and through (s,a) after t simulations, and $\overline{Q}_t(s,a)$ is the empirical mean of terminal returns from simulations passing through (s,a). Unvisited actions receive infinite score. Rewards lie in $[-1,1]$, so (17) applies the UCB1 bonus for a unit-range reward to a range-2 one; the constant is held fixed across all conditions rather than tuned, so it cannot by itself explain a difference between them. Each simulation selects through the current tree, expands one unvisited edge, performs one uniform random completion, and backs up the terminal reward by an arithmetic mean.

Max-UCT diagnostic. Define

$$Q_t^{\max}(s,a) = \max\left\{R(\pi_b) : \pi_b \text{ was sampled through } (s,a)\right\}. \tag{18}$$

Select using

$$a_t = \arg\max_a \left[Q_t^{\max}(s,a) + \sqrt{\frac{2\log N_t(s)}{N_t(s,a)}} \right]. \tag{19}$$

The tree structure, expansion policy, rollout policy, and exploration term are otherwise identical to Mean-UCT. This condition is a causal diagnostic of the mean-versus-maximum distinction, not a claim that naive maximum backup is universally optimal. A calibrated extreme-bandit allocation rule is a natural subsequent comparison [4].

5.2 Evaluation protocol

Each search run uses one persistent tree rooted at s_0. The algorithm does not commit to a grammar action before the budget is exhausted. No network, learned prior, Dirichlet noise, reward clipping, complexity penalty, or nonlinear optimizer is used. One simulation contains exactly one terminal reward evaluation.

Use the terminal-evaluation budgets

$$\mathcal{B} = \{16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192\}. \tag{20}$$

For each of the four benchmark cells and each search condition, use 100 independent seeds. Run each seed once to $B_{\max} = 8192$ and record all intermediate checkpoints. The same base-seed list is used across algorithms.

5.3 Outcomes

Let $\pi_1, \ldots, \pi_B$ be the terminal expressions observed by budget B. Define the best-observed reward

$$R_B^{\max} = \max_{1 \le b \le B} R(\pi_b), \tag{21}$$

the simple regret

$$r_B = 1 - R_B^{\max}, \tag{22}$$

and the exact-recovery indicator

$$Z_B = \mathbf{1}\left\{R_B^{\max} \ge 1 - \tau\right\}. \tag{23}$$

The budget at which exact recovery first occurs is

$$B_{\text{exact}} = \inf\{B : Z_B = 1\}, \tag{24}$$

written with Z and B rather than S and T, which Section 1.3 reserves for the start symbol and the transition map.

Exact recovery is the primary outcome. Simple regret is secondary because a structurally wrong expression already achieves small regret: on `quad_B` the plateau of Section 3.1 sits at

$$1 - 0.997213 = 0.002787,$$

and on `lin_A` the decoy D_2 = `* C1 x` sits at $1 - 0.821429 = 0.178571$ without containing a constant term at all. Thus predictive reward alone cannot distinguish near-perfect approximation from structural recovery. This distinction is consistent with symbolic regression benchmarks that report both predictive and exact-recovery performance [7].

For mechanism diagnosis, record the continue-visit fraction

$$F_B = \frac{N_B(s_0, C)}{N_B(s_0)}, \tag{25}$$

where $N_B(s_0)$ and $N_B(s_0, C)$ are the root and continue-edge visit counts at budget B, together with the final root recommendation, number of distinct terminal sequences, number of distinct canonical atom sets, and the best reward found through each root action. The final visit-count recommendation is diagnostic only; the primary search output is the best terminal expression encountered.

5.4 Hypotheses

Hypothesis 2a: value-semantic effect. Mean-UCT will allocate less search to the continue branch on the inversion cells (`lin_A`, `quad_D`) than on their non-inversion partners (`lin_B`, `quad_A`), because its exploitation statistic is anchored near $V^q(C)$ while subtree sampling remains close to uniform, and $V^q(C)$ lies below $\max_i R(D_i)$ exactly on those cells. Max-UCT will be less sensitive to the inversion after observing high-reward continuations.

Hypothesis 2b: rarity effect. For every algorithm, exact recovery will be slower on the quadratic cells ($\rho(C) = 0.11$) than on the linear ones (0.39). At moderate budgets, recovery will be organized approximately by the effective rare-event budget

$$B\rho. \tag{26}$$

Hypothesis 2c: interaction. The largest recovery difference between Max-UCT and Mean-UCT will occur on `quad_D`, the inversion-and-rare cell, where the continue branch has both a lower rollout mean than a one-action terminal and the small exact mass $\rho(C) = 0.1128$.

One pre-specified interaction contrast is

$$\begin{aligned}\Delta_{\text{semantic}}(B) = \big[P_B^{\text{Max}} - P_B^{\text{Mean}}\big]_{\text{inversion}} \\ - \big[P_B^{\text{Max}} - P_B^{\text{Mean}}\big]_{\text{no inversion}},\end{aligned} \tag{27}$$

where P_B^{Max} and P_B^{Mean} are the exact-recovery probabilities $\Pr(Z_B = 1)$ of Max-UCT and Mean-UCT, each bracket averaged over the two cells at that level of the design. The prediction is

$$\Delta_{\text{semantic}}(B) > 0,$$

with a larger effect expected on the quadratic (rare) cells.

5.5 Statistical reporting and interpretation

For each budget, cell, and algorithm, report the exact-recovery proportion with a 95% Wilson interval, median simple regret with a seed-bootstrap interval, the empirical survival curve

$$\Pr(B_{\text{exact}} > B),$$

and the continue-visit fraction F_B.

The interpretation is mechanism-based:

- If Mean-UCT fails while Max-UCT succeeds, the mean-return statistic is causally implicated.
- If both methods deteriorate primarily on the quadratic cells, completion rarity is the dominant limitation.
- If random search outperforms Mean-UCT on an inversion cell, the tree allocation rule actively suppresses the optimal branch.
- If Mean-UCT succeeds despite the exact value-order inversion, UCT exploration is sufficient at the tested budgets and the mismatch alone is not a complete explanation.
- If Max-UCT performs worse on the non-inversion cells, maximum backup is overoptimistic and should not be treated as a universal replacement for mean backup.
- If simple regret is small while exact recovery remains poor, the experiment demonstrates a separation between predictive approximation and symbolic identification.

5.6 Secondary validation and deferred questions

After the two primary experiments, a nonconfirmatory validation may repeat the key comparison on the full seven-production grammar, whose sine and division productions are pure distractors for these targets: at $L = 12$ it has 795,437 reachable forms and 19,438 terminals against the 4,898 and 247 used above, so rarity there is a grammar effect rather than a target effect. Nonlinear constant fitting, uniform-prior PUCT, learned policy and value heads, and SRBench evaluation enter only after the controlled mechanisms are understood. In particular, whether a learned policy or value function corrects the pure-search failures identified here — or instead learns and amplifies the biased targets produced by mean-return search — is deferred, so that the present study remains causally interpretable.

References

[1] Tristan Cazenave. Nested monte-carlo search. In *Proceedings of the 21st International Joint Conference on Artificial Intelligence*, pages 456–461, 2009.

[2] Tristan Cazenave. Monte-carlo expression discovery. *International Journal on Artificial Intelligence Tools*, 22(1):1250035, 2013.

[3] Pierre-Arnaud Coquelin and Rémi Munos. Bandit algorithms for tree search. In *Proceedings of the Twenty-Third Conference on Uncertainty in Artificial Intelligence*, pages 67–74. AUAI Press, 2007.

[4] Zhengyao Huang, Daniel Zhengyu Huang, Tiannan Xiao, Dina Ma, Zhenyu Ming, Hao Shi, and Yuanhui Wen. Improving monte carlo tree search for symbolic regression. In *Advances in Neural Information Processing Systems*, volume 38, 2025.

[5] Pierre-Alexandre Kamienny, Guillaume Lample, Sylvain Lamprier, and Marco Virgolin. Deep generative symbolic regression with Monte-Carlo-tree-search. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett, editors, *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 15655–15668. PMLR, 2023.

[6] Levente Kocsis and Csaba Szepesvári. Bandit based monte-carlo planning. In *Machine Learning: ECML 2006*, volume 4212 of *Lecture Notes in Computer Science*, pages 282–293. Springer, 2006.

[7] William La Cava, Patryk Orzechowski, Bogdan Burlacu, Fabrício Olivetti de França, Marco Virgolin, Ying Jin, Michael Kommenda, and Jason H. Moore. Contemporary symbolic regression methods and their relative performance. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, volume 1, 2021.

[8] Brenden K. Petersen, Mikel Landajuela, T. Nathan Mundhenk, Claudio P. Santiago, Soo K. Kim, and Joanne T. Kim. Deep symbolic regression: Recovering mathematical expressions from data via risk-seeking policy gradients. In *International Conference on Learning Representations*, 2021.